

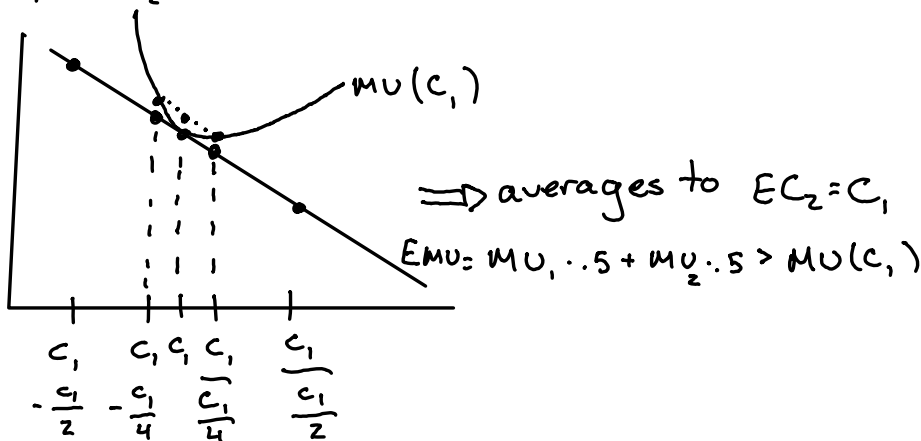
01-13-2025

Saving with uncertainty (Precautionary):

$$\hookrightarrow C_2 = C_1 + \begin{cases} c_1/4 & ; P = \frac{1}{2} \\ -c_1/4 & ; P = \frac{1}{2} \end{cases}$$

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$$C_1 = EC_2 \Rightarrow MU(C_1) \text{ vs. } EMU(C_2)$$



$\rightarrow$ eliminate cons. gap in Euler by saving
 $\hookrightarrow$ uncertainty $\uparrow$; savings $\uparrow$

Motive to Save 4: Bequest:

$$\hookrightarrow \max_{C_1, C_2, a_3} U(C_1) + \beta U(C_2) + \underbrace{\beta^2 \gamma(a_3)}_{\text{bequest motive}}$$

$$\text{s.t. } C_1 = y_1 - a_2$$

$$C_2 = y_2 + a_2(1+r) - a_3$$

bequest motive

Euler Equation Refresher:

$$\hookrightarrow \max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

s.t.

$$c_t = k_t(1+r) + y_t - k_{t+1}; \lambda_t; \beta^t$$

$$\mathcal{L} = E_0 \left\{ \sum_{t=0}^{\infty} \beta^t (u(c_t) + \lambda_t (k_t(1+r) + y_t - k_{t+1} - c_t)) \right\}$$

FOC:

$$\hookrightarrow c_t: u'(c_t) = \lambda_t$$

$$\hookrightarrow k_{t+1}: -\beta^t \lambda_t + \beta^{t+1} (1+r) \lambda_{t+1} = 0$$

$$\rightarrow \beta(1+r) E_t \lambda_{t+1} = \lambda_t$$

$$\rightarrow u'(c_t) = \beta(1+r) E_t u'(c_{t+1})$$

Permanent Income Hypothesis:

$\hookrightarrow$ Assume no uncertainty: $t=0, \dots, T$; $\beta(1+r)=1$

$$\rightarrow \max_{\{c_t\}_{t=0}^{\infty}} \sum_{t=0}^T \beta^t u(c_t)$$

$$\{a_{t+1}\}_{t=0}^T \quad \sum_{t=0}^T \frac{c_t}{(1+r)^t} = a_0 + \sum_{t=0}^T \frac{y_t}{(1+r)^t}$$

s.t.

$\hookrightarrow$ Euler:

$$\rightarrow u'(c_t) = \beta(1+r) u'(c_{t+1}) \Rightarrow u'(c_t) = u'(c_{t+1})$$

due to assump.

Permanent Income Hypothesis:

$$\hookrightarrow \therefore C_t = C_{t+1} = C \quad \forall t$$

$$\rightarrow C \sum_{t=0}^T \frac{1}{(1+r)^t} = a_0 + \sum_{t=0}^T \frac{y_t}{(1+r)^t}$$

$$\rightarrow \text{Geom Series: } C \cdot \frac{1 - (\frac{1}{1+r})^{T+1}}{1 - \frac{1}{1+r}} = a_0 + \sum_{t=0}^T \frac{y_t}{(1+r)^t}$$

$\hookrightarrow$ Review formula

$$\rightarrow C = \frac{1 - \frac{1}{(1+r)}}{1 - (\frac{1}{1+r})^{T+1}} \cdot \left[a_0 + \sum_{t=0}^T \frac{y_t}{(1+r)^t} \right]$$

$$= \underbrace{\frac{1-\beta}{1-\beta^{T+1}}}_{\beta = \frac{1}{1+r}} \underbrace{\left[a_0 + \sum_{t=0}^T \frac{y_t}{(1+r)^t} \right]}_{\text{lifetime wealth}}$$

$\rightarrow a_0$: financial wealth; $\sum_{t=0}^T \frac{y_t}{(1+r)^t}$: human wealth

$\hookrightarrow C = \text{permanent income [Friedman]}$

What is PR (Friedman)? : Annuity value of Total Wealth

$\hookrightarrow$ annuity: pay in and receive until death

$$\rightarrow q = \sum_{t=0}^T \frac{1}{(1+r)^t} \quad (\text{price of annuity})$$

$$\hookrightarrow q = \frac{1 - (\frac{1}{1+r})^{T+1}}{1 - (\frac{1}{1+r})}; \text{ pay in } W \Rightarrow \text{ receive } \frac{W}{q} \quad \forall t, \dots, T$$

Permanent Income (Friedman)

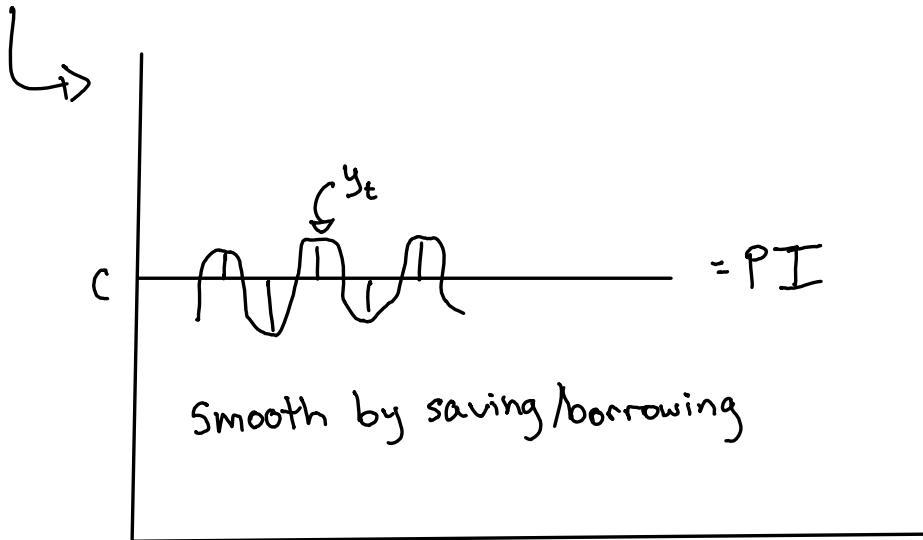
↳ Infinite horizon case:

$$\rightarrow \frac{1 - (\frac{1}{1+r})^{T+1}}{1 - (\frac{1}{1+r})} = \frac{1}{1 - (\frac{1}{1+r})} = \frac{1+r}{r} = \frac{1}{\beta} = \frac{1}{1-\beta}$$

$$C = (1-\beta) \left[a_0 + \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} \right]$$

↳ Formally:

→ Consumption is not determined by income in that period but by Lifetime Wealth or Permanent Income.



M-B Life Cycle Model (1954):

↳ $1 \rightarrow T, 1, \dots, N$: Working; $N+1:T$: Retired.

→ Assump: $\beta=1$; $r=0$

→ Euler: $U'(c_t) = U'(c_{t+1}) \Rightarrow c = c_t = c_{t+1}$